BOYA AJAY VARMA

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GitHub: github.com/ajayvarmaboya | LeetCode: leetcode.com/u/ajayvarmab | Portfolio: ajayvarmaboya.github.io/portfolio

Junior AI & Backend Engineer | Python, Java, FastAPI, Spring Boot & Agentic RAG

PROFESSIONAL SUMMARY

Computer Science (AI & ML) graduate specializing in backend engineering and document-grounded GenAI architectures. Hands-on experience designing high-throughput REST APIs with FastAPI and Spring Boot, building stateful LangGraph RAG workflows, and implementing layered DAO architectures with PostgreSQL. Proven problem-solver with open-source contributions and active algorithmic practice in Java and Python.

TECHNICAL SKILLS

- **Programming Languages:** Java (17/21), Python, JavaScript, SQL
- **Backend & APIs:** FastAPI, Spring Boot, RESTful APIs, JDBC, Pydantic, Uvicorn, OAuth 2.0
- **AI & Vector Search:** LangGraph, LangChain, Pinecone, FAISS, Sentence Transformers, RAG Workflows, Prompt Engineering
- **Databases & Storage:** PostgreSQL, MySQL, Schema Design, Relational Modeling, PreparedStatement
- **DevOps & Cloud:** Docker, Docker Compose, Git, GitHub Actions, Linux, CI/CD Fundamentals, AWS & OCI Fundamentals
- **Core Engineering:** OOP, Data Structures & Algorithms (DSA), DAO Pattern, Layered Architecture, Exception Handling

SELECTED ENGINEERING PROJECTS

Legixo RAG — Document-Grounded GenAI Pipeline

Technologies: Python, FastAPI, LangGraph, Pinecone, Groq LLM, Sentence Transformers, Pydantic

- Architected a document-grounded question-answering service using LangGraph StateGraph with conditional routing between retrieval relevance filtering, source attribution, and response synthesis.
- Integrated Pinecone vector database with Sentence Transformers embeddings for semantic chunk retrieval, verifying 12/12 automated evaluation test scenarios for hallucination reduction.
- Exposed asynchronous REST endpoints via FastAPI and Uvicorn with strict Pydantic schema validation for request payloads and structured responses.

LeetCode GitHub Sync — Browser Extension & Microservice

Technologies: Java, Spring Boot, JavaScript, Manifest V3, GitHub REST API, OAuth 2.0, Maven

- Engineered an automated browser extension detecting accepted LeetCode submissions and committing structured solution files directly to user GitHub repositories via GitHub REST API.
- Implemented secure GitHub OAuth token exchange through a Spring Boot backend service, integrating Monaco editor DOM extraction and UTF-8-safe Base64 encoding.
- Built validation logic to eliminate duplicate commits, handle wrong-answer protection, and provide structured sync error logging.

JobPrep OS & StockFlow Backend

Technologies: Java 17, Spring Boot 3, PostgreSQL, Docker Compose, Maven, Relational Modeling

- Designed relational database schemas and indexed PostgreSQL tables to support interview preparation tracking and order/inventory management workflows.
- Containerized application and database services using Docker Compose, establishing clean environment configuration and reliable local database persistence.
- Implemented presentation, service, and DAO layers with constructor dependency injection, robust input validation, and custom exception handling.

OpenWork — Open-Source Agentic AI Application

Technologies: Open Source, Docker, Git/GitHub, Linux, Agentic Workflows

- Contribute to an open-source desktop application powering guided agentic workflows, participating in PR reviews, issue triage, and code contributions.
- Configured and validated multi-container Docker environments for running local autonomous agents and backend execution tasks.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML)

Sri Venkateswara Engineering College, Tirupati, India | Graduation: 2025 | CGPA: **8.2 / 10**

CERTIFICATIONS & KEY HONORS

- **NPTEL National Certification:** Fundamentals of Cloud, Edge, IoT & ML — **Top 5% National Rank**
- **Algorithmic Practice:** Active problem-solving in Java & Python with structured solutions in public *neetcode-submissions* repository.